General Tolerance Chart

ISO 2768 1 & 2 | ISO 286

Edited by ChansMachining

ISO system of limits and fits **fundamental tolerances**																			
nominal dimension in mm		micrometer (1/1000 mm)											millimeter						
over	up to	IT1	IT2	IT3	IT4	IT5	IT6	IT7	IT8	IT9	IT10	IT11	IT12	IT13	IT14	IT15	IT16	IT17	IT18
-	3	0.8	1.2	2	3	4	6	10	14	25	40	60	0.1	0.14	0.25	0.4	0.6	1	1.4
3	6	1	1.5	2.5	4	5	8	12	18	30	48	75	0.12	0.18	0.3	0.48	0.75	1.2	1.8
6	10	1	1.5	2.5	4	6	9	15	22	36	58	90	0.15	0.22	0.36	0.58	0.9	1.5	2.2
10	18	1.2	2	3	5	8	11	18	27	43	70	110	0.18	0.27	0.43	0.7	1.1	1.8	2.7
18	30	1.5	2.5	4	6	9	13	21	33	52	84	130	0.21	0.33	0.52	0.84	1.3	2.1	3.3
30	50	1.5	2.5	4	7	11	16	25	39	62	100	160	0.25	0.39	0.62	1	1.6	2.5	3.9
50	80	2	3	5	8	13	19	30	46	74	120	190	0.3	0.46	0.74	1.2	1.9	3	4.6
80	120	2.5	4	6	10	15	22	35	54	87	140	220	0.35	0.54	0.87	1.4	2.2	3.5	5.4
120	180	3.5	5	8	12	18	25	40	63	100	160	250	0.4	0.63	1	1.6	2.5	4	6.3
180	250	4.5	7	10	14	20	29	46	72	115	185	290	0.46	0.72	1.15	1.85	2.9	4.6	7.2
250	315	6	8	12	16	23	32	52	81	130	210	320	0.52	0.81	1.3	2.1	3.2	5.2	8.1
315	400	7	9	13	18	25	36	57	89	140	230	360	0.57	0.89	1.4	2.3	3.6	5.7	8.9
400	500	8	10	15	20	27	40	63	97	155	250	400	0.63	0.97	1.55	2.5	4	6.3	9.7
500	630	9	11	16	22	32	44	70	110	175	280	440	0.7	1.1	1.75	2.8	4.4	7	11
630	800	10	13	18	25	36	50	80	125	200	320	500	0.8	1.25	2	3.2	5	8	12.5
800	1000	11	15	21	28	40	56	90	140	230	360	560	0.9	1.4	2.3	3.6	5.6	9	14
1000	1250	13	18	24	33	47	66	105	165	260	420	660	1.05	1.65	2.6	4.2	6.6	10.5	16.5
1250	1600	15	21	29	39	55	78	125	195	310	500	780	1.25	1.95	3.1	5	7.8	12.5	19.5
1600	2000	18	25	35	46	65	92	150	230	370	600	920	1.5	2.3	3.7	6	9.2	15	23
2000	2500	22	30	41	55	78	110	175	280	440	700	1100	1.75	2.8	4.4	7	11	17.5	28
2500	3150	26	36	50	68	96	135	210	330	540	860	1350	2.1	3.3	5.4	8.6	13.5	21	33
do not use IT14 to IT18 for nominal dimensions of 1 mm and below																			

CHANS
MACHINING PARTS

ISO system of limits and fits(basic shaft system)

upper and lower limits in micrometer (0.001 mm)

over	up to	d9	e8	f7	g6	h5	h6	h7	h8	h9	h11	js5	js6	js13	js14	k5	k6	m5	m6	n5	n6	p6	r6	s6	s7
-	3	-20 -45	-14 -28	-6 -16	-2 -8	0 -4	0 -6	0 -10	0 -14	0 -25	0 -60	2 -2	3 -3	70 -70	125 -125	4 0	6 0	6 2	8 2	8 4	10 4	12 6	16 10	20 14	24 14
3	6	-30 -60	-20 -38	-10 -22	-4 -12	0 -5	0 -8	0 -12	0 -18	0 -30	0 -75	2.5 -2.5	4 -4	90 -90	150 -150	6 1	9 1	9 4	12 4	13 8	16 8	20 12	23 15	27 19	31 19
6	10	-40 -76	-25 -47	-13 -28	-5 -14	0 -6	0 -9	0 -15	0 -22	0 -36	0 -90	3 -3	4.5 -4.5	110 -110	180 -180	7 1	10 1	12 6	15 6	16 10	19 10	24 15	28 19	32 23	38 23
10	18	-50 -93	-32 -59	-16 -34	-6 -17	0 -8	0 -11	0 -18	0 -27	0 -43	0 -110	4 -4	5.5 -5.5	135 -135	215 -215	9 1	12 1	15 7	18 7	20 12	23 12	29 18	34 23	39 28	46 28
18	30	-65 -117	-40 -73	-20 -41	-7 -20	0 -9	0 -13	0 -21	0 -33	0 -52	0 -130	4.5 -4.5	6.5 -6.5	165 -165	260 -260	11 2	15 2	17 8	21 8	24 15	28 15	35 22	41 28	48 35	56 35
30	50	-80 -142	-50 -89	-25 -50	-9 -25	0 -11	0 -16	0 -25	0 -39	0 -62	0 -160	5.5 -5.5	8 -8	195 -195	310 -310	13 2	18 2	20 9	25 9	28 17	33 17	42 26	50 34	59 43	68 43
50	65	-100 -174	-60 -106	-30 -60	-10 -29	0 -13	0 -19	0 -30	0 -46	0 -74	0 -190	6.5 -6.5	9.5 -9.5	230 -230	370 -370	15 2	21 2	24 11	30 11	33 20	39 20	51 32	60 41	72 53	83 53
65	80	-100 -174	-60 -106	-30 -60	-10 -29	0 -13	0 -19	0 -30	0 -46	0 -74	0 -190	6.5 -6.5	9.5 -9.5	230 -230	370 -370	15 2	21 2	24 11	30 11	33 20	39 20	51 32	62 43	78 59	89 59
80	100	-120 -207	-72 -126	-36 -71	-12 -34	0 -15	0 -22	0 -35	0 -54	0 -87	0 -220	7.5 -7.5	11 -11	270 -270	435 -435	18 3	25 3	28 13	35 13	38 23	45 23	59 37	73 51	93 71	106 71
100	120	-120 -207	-72 -126	-36 -71	-12 -34	0 -15	0 -22	0 -35	0 -54	0 -87	0 -220	7.5 -7.5	11 -11	270 -270	435 -435	18 3	25 3	28 13	35 13	38 23	45 23	59 37	76 54	101 79	114 79
120	140	-145 -245	-85 -148	-43 -83	-14 -39	0 -18	0 -25	0 -40	0 -63	0 -100	0 -250	9 -9	12.5 -12.5	315 -315	500 -500	21 3	28 3	33 15	40 15	45 27	52 27	68 43	88 63	117 92	132 92
140	160																						90 65	125 100	140 100
160	180																						93 68	133 108	148 108
180	200	-170 -285	-100 -172	-50 -96	-15 -44	0 -20	0 -29	0 -46	0 -72	0 -115	0 -290	10 -10	14.5 -14.5	360 -360	575 -575	24 4	33 4	37 17	46 17	51 31	60 31	79 50	106 77	151 122	168 122
200	225																						109 80	159 130	176 130
225	250																						113 84	169 140	186 140
250	280	-190 -320	-110 -191	-56 -108	-17 -49	0 -23	0 -32	0 -52	0 -81	0 -130	0 -320	11.5 -11.5	16 -16	405 -405	650 -650	27 4	36 4	43 20	52 20	57 34	66 34	88 56	126 94	190 158	210 158
280	315																						130 98	202 170	222 170
315	355	-210 -350	-125 -214	-62 -119	-18 -54	0 -25	0 -36	0 -57	0 -89	0 -140	0 -360	12.5 -12.5	18 -18	445 -445	700 -700	29 4	40 4	46 21	57 21	62 37	73 37	98 62	144 108	226 190	247 190
355	400																						150 114	244 208	265 208

ISO system of limits and fits(basic hole system)

upper and lower limits in micrometer (0.001 mm)

over	up to	D10	E9	F7	F8	G7	G9	H6	H7	H8	H9	H11	H12	H13	JS7	JS9	K6	K7	M6	M7	N7	N9	P7	P9	R7
-	3	60 20	39 14	16 6	20 10	12 2	27 2	6 0	10 0	14 0	25 0	60 0	100 0	140 0	5 -5	12.5 -12.5	0 -6	0 -10	-2 -8	-2 -12	-4 -14	-4 -29	-6 -16	-6 -31	-10 -20
3	6	78 30	50 20	22 10	28 10	16 4	34 4	8 0	12 0	18 0	30 0	75 0	120 0	180 0	6 -6	15 -15	2 -6	3 -9	-1 -9	0 -12	-4 -16	0 -30	-8 -20	-12 -42	-11 -23
6	10	98 40	61 25	28 13	35 13	20 5	41 5	9 0	15 0	22 0	36 0	90 0	150 0	220 0	7.5 -7.5	18 -18	2 -7	5 -10	-3 -12	0 -15	-4 -19	0 -36	-9 -24	-15 -51	-13 -28
10	18	120 50	75 32	34 16	43 16	24 6	49 6	11 0	18 0	27 0	43 0	110 0	180 0	270 0	9 -9	21.5 -21.5	2 -9	6 -12	-4 -15	0 -18	-5 -23	0 -43	-11 -29	-18 -61	-16 -34
18	30	149 65	92 40	41 20	53 20	28 7	59 7	13 0	21 0	33 0	52 0	130 0	210 0	330 0	10.5 -10.5	26 -26	2 -11	6 -15	-4 -17	0 -21	-7 -28	0 -52	-14 -35	-22 -74	-20 -41
30	50	180 80	112 50	50 25	64 25	34 9	71 9	16 0	25 0	39 0	62 0	160 0	250 0	390 0	12.5 -12.5	31 -31	3 -13	7 -18	-4 -20	0 -25	-8 -33	0 -62	-17 -42	-26 -88	-25 -50
50	65	220 100	134 60	60 30	76 30	40 10		19 0	30 0	46 0	74 0	190 0	300 0	460 0	15 -15	37 -37	4 -15	9 -21	-5 -24	0 -30	-9 -39	0 -74	-21 -51	-32 -106	-30 -60
65	80	220 100	134 60	60 30	76 30	40 10		19 0	30 0	46 0	74 0	190 0	300 0	460 0	15 -15	37 -37	4 -15	9 -21	-5 -24	0 -30	-9 -39	0 -74	-21 -51	-32 -106	-32 -62
80	100	260 120	159 72	71 36	90 36	47 12		22 0	35 0	54 0	87 0	220 0	350 0	540 0	17.5 -17.5	43.5 -43.5	4 -18	10 -25	-6 -28	0 -35	-10 -45	0 -87	-24 -59	-37 -124	-38 -73
100	120	260 120	159 72	71 36	90 36	47 12		22 0	35 0	54 0	87 0	220 0	350 0	540 0	17.5 -17.5	43.5 -43.5	4 -18	10 -25	-6 -28	0 -35	-10 -45	0 -87	-24 -59	-37 -124	-41 -76
120	140	305 145	185 85	83 43	106 43	54 14		25 0	40 0	63 0	100 0	250 0	400 0	630 0	20 -20	50 -50	4 -21	12 -28	-8 -33	0 -40	-12 -52	0 -100	-28 -68	-43 -143	-48 -88
140	160																								-50 -90
160	180																								-53 -93
180	200	335 170	215 110	96 50	122 50	61 15		29 0	46 0	72 0	115 0	290 0	460 0	720 0	23 -23	57.5 -57.5	5 -24	13 -33	-8 -37	0 -46	-14 -60	0 -115	-33 -79	-50 -165	-60 -106
200	225																								-63 -109
225	250																								-67 -113
250	280	400 190	240 110	108 56	137 56	69 17		32 0	52 0	81 0	130 0	320 0	520 0	810 0	26 -26	65 -65	5 -27	16 -36	-9 -41	0 -52	-14 -66	0 -130	-36 -88	-56 -186	-74 -126
280	315																								-78 -130
315	355	440 210	265 125	119 62	151 62	75 18		36 0	57 0	89 0	140 0	360 0	570 0	890 0	28.5 -28.5	70 -70	7 -29	17 -40	-10 -46	0 -57	-16 -73	0 -140	-41 -98	-62 -202	-87 -144
355	400																								-93 -150

do not use N9 for nominal dimensions of 1 mm or below

basic hole	clearance fits	basic shaft
H8/d9	loose running fit: clearance allows for loose fit of parts	D10/h
H8/e8	free running fit: sufficient clearance is allowed for ease of assembly	E9/h9
H8/f7	close running fit: clearance allows for parts to be easily assembled by hand while maintaining location accuracy	F8/h9
H7/f7	sliding fit - free: clearance allows accurate location and free movement, including turning	F8/h6
H7/g6	sliding fit - constrained: cleaance allows better location accuracy while still allowing sliding or turning movement	G7/h6
H8/h9	minimal clearance fit: allows location accuracy and hand force assembly without being a snug fit	H8/h9
H7/h6	locational clearance fit: Allows snug fit of stationary parts that may be assembled by hand force	H7/h6
	transition fits	
H7/j6	locational transition fit: for accurate location allowing more clearance than interference	not specifi
H7/n6	locational transition fit - interference: For accurate location where interference is permissible	
	interference fits	
H7/r6	locational interference fit: for rigidity and alignment/accurate location without special bore requirements	not specifi ed
H7/s6	medium drive fit: for ordinary steel parts or shrink fits of light sections, tightest fit possible for cast iron	
H8/u8	force fit: for parts fitting that can withstand high mechanical pressing force or shrink fitting	
H8/x8	extreme force fit: for parts that can only be assembled by stretching or shrinking	

General tolerances for linear and angular dimensions

tolerances for cutting and forming processes

	linear dimensions mm							
tolerance class	0.5 to 3	over 3 to 6	over 6 to 30	over 30 to 120	over 120 to 400	over 400 to 1000	over 1000 to 2000	over 2000 to 4000
f (fine)	+/- 0.05	+/- 0.05	+/- 0.1	+/- 0.15	+/- 0.2	+/- 0.3	+/- 0.5	-
m (medium)	+/- 0.1	+/- 0.1	+/- 0.2	+/- 0.3	+/- 0.5	+/- 0.8	+/- 1.2	+/- 2.0
c (coarse)	+/- 0.2	+/- 0.3	+/- 0.5	+/- 0.8	+/- 1.2	+/- 2.0	+/- 3.0	+/- 4.0
v (very coarse)	-	+/- 0.5	+/- 1.0	+/- 1.5	+/- 2.5	+/- 4.0	+/- 6.0	+/- 8.0

	radii and chamfers			**angular dimensions** (nominal dimensions for the shorter angle leg)				
	0.5 to 3	over 3 to 6	over 6	up to 10	over 10 to 50	over 50 to 120	over 120 to 400	over 400
f (fine) m (medium)	+/- 0.2	+/- 0.5	+/- 1.0	+/- 1°	+/- 0° 30'	+/- 0° 20'	+/- 0° 10'	+/- 0° 5'
c (coarse)	+/- 0.4	+/- 1.0	+/- 2.0	+/- 1° 30'	+/- 1°	+/- 0° 30'	+/- 0° 15'	+/- 0° 10'
v (very coarse)				+/- 3°	+/- 2°	+/- 1°	+/- 0° 30'	+/- 0° 20'

General tolerances for shape and position

	straightness and flatness(nominal dimensions in mm)						**circular runout**
	up to 10	over 10 to	over 30 to	over 100 to 300	over 300 to	over 1000 to 3000	
H	0.02	0.05	0.1	0.2	0.3	0.4	0.1
K	0.05	0.1	0.2	0.4	0.6	0.8	0.2
L	0.1	0.2	0.4	0.8	1.2	1.6	0.5

	perpendicularity (nominal dimensions for the shorter leg)				**symmetry**			
	up to 100	over 100 to 300	over 300 to 1000	over 1000 to 3000	up to 100	over 100 to 300	over 300 to 1000	over 1000 to 3000
H	0.2	0.3	0.4	0.5	0.5	0.5	0.5	0.5
K	0.4	0.6	0.8	1	0.6	0.6	0.8	1
L	0.6	1	1.5	2	0.6	1	1.5	2